

SENZO NCEKANA

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DATA SCIENTIST / DATA ANALYST — RISK, AUDIT & COMPLIANCE ANALYTICS

Data analytics and data science professional delivering coding-driven analytics in Python and SQL — building reusable, automated analytics, models and dashboards from raw, multi-source data. Strong in ETL/data preparation, applied machine learning, and anomaly/exception & pattern detection across full data populations. 3+ years in regulated healthcare/pharmacy (compliance, control adherence, data integrity for audits), now focused on assurance, controls, and a move into banking and financial services. Comfortable communicating insight to technical and non-technical stakeholders.

TECHNICAL SKILLS

Programming & Data: Python, SQL, SQLite, R, Relational Databases, Data Warehousing Concepts, ETL / Data Preparation, Git, Django, Docker.

Analytics & Data Science: Applied Machine Learning (Supervised & Unsupervised), Anomaly / Outlier & Exception Detection, Clustering, Classification, Feature Engineering, EDA, Statistical Analysis, NLP.

BI, Visualization & Reporting: Power BI, Tableau, Excel (Pivot Tables, VLOOKUP, Macros), Dashboarding, Data Dictionaries, Reporting, Data Visualization, Trend Analysis.

Soft Skills: Stakeholder Engagement · Trusted Advisor & Influencing · Analytical Thinking · Attention to Detail · Problem-Solving · Managing Multiple Priorities · Comfortable with Ambiguity / Agile.

EXPERIENCE

AI Internship | CodeAlpha 03/2026 – 04/2026

- Built a question-matching system using cosine similarity to surface the most relevant records — transferable to data matching, deduplication, and reconciliation in audit analytics.
- Developed an NLP FAQ chatbot (NLTK, SpaCy) with text preprocessing and intent matching; also implemented computer-vision (YOLO, OpenCV) and LSTM sequence models.

AI Engineering Bootcamp | HyperionDev 02/2025 – 01/2026

- Built end-to-end analytics solutions integrating data ingestion, cleaning, transformation (ETL), and predictive modeling.
- Designed reusable, reproducible workflows for structured and unstructured data, emphasizing accuracy and data integrity.
- Developed a full-stack Django app with SQL / relational database design; conducted EDA and built dashboards to turn data into decision-ready insight.
- Applied supervised/unsupervised ML, neural networks, and NLP; used Git/GitHub for version control and collaboration.

Pharmacy Technician | Nelson Mandela Children's Hospital 07/2023 – Present

- Maintain accurate electronic records and inventory data, ensuring data integrity for reporting and audits in a regulated environment.
- Operate within strict regulatory standards, contributing to control adherence and control-effectiveness in a high-volume dispensary.
- Monitor stock and consumption data to detect exceptions and anomalies, supporting formulary and stock-control decisions.
- Analyze workflow data to drive process improvements; collaborate across technical and non-technical stakeholders (doctors, nurses, pharmacists).

Pharmacy Technician | Local Choice Pelican Pharmacy 01/2022 – 06/2023

- Ensured accurate data entry, record-keeping, and data integrity across dispensary operations while maintaining regulatory compliance.
- Monitored stock trends to flag exceptions and reduced dispensing errors through process improvements.

Pharmacy Technician Intern | Prince Mshiyeni Memorial Hospital 04/2021 – 11/2021

- Monitored inventory levels and prepared reporting to support audits and regulatory compliance.
- Documented patient and stock data accurately; collaborated with healthcare staff to optimize medication processes.

Lab Assistant (Manufacturing) | Nelson Mandela University 06/2019 – 11/2019

- Collected, recorded, and analyzed production data to identify efficiency improvements.
- Performed quality-control checks to ensure compliance with safety and quality standards in regulated manufacturing.

PROJECTS

Customer Churn Prediction – Risk Scoring ML Pipeline

- Built an end-to-end ML pipeline to score and predict customer churn (EDA, feature engineering, model comparison, hyperparameter tuning, ROC-AUC/F1 evaluation) — risk/exception scoring of a population, analogous to flagging high-risk items for control testing.

Tech: Python, pandas, NumPy, scikit-learn (Logistic Regression, Random Forest, Gradient Boosting), XGBoost, Matplotlib, Seaborn. · [Repo](#)

US Crime Statistics Analyst – Pattern & Anomaly Detection

- Segmented and detected patterns across a full state-level population using PCA and K-means/Hierarchical clustering, with statistical interpretation and visualization — directly analogous to full-population analysis and outlier/exception detection in audit.

Tech: Python, pandas, NumPy, scikit-learn, SciPy, Matplotlib, Seaborn. · [Repo](#)

PharmaGuard – Pharmaceutical Similarity Detection & Compliance Platform

- Built an end-to-end AI platform detecting look-alike sound-alike (LASA) risks in pharmaceutical names/packaging, supporting regulatory compliance-review workflows — an AI-powered approach to assurance in a regulated industry.
- Designed a multi-agent pipeline (LangChain, Google Gemini) with RAG (FAISS) and OCR/computer vision; delivered a full-stack FastAPI + Next.js app, Dockerized with GitHub Actions CI/CD.

Tech: Python, FastAPI, Next.js, TypeScript, Google Gemini, LangChain, FAISS, PaddleOCR, PostgreSQL, Docker. · [Repo](#)

Amazon Review Sentiment Analysis – Unstructured-Data Analytics at Scale

- Built an NLP pipeline over 28,000+ reviews (preprocessing, polarity classification, sentiment scoring, similarity analysis) with reproducible workflows — turning large volumes of unstructured data into interpretable insight.

Tech: Python, spaCy, SpacyTextBlob, Jupyter Notebook, Data Cleaning, Text Preprocessing. · [Repo](#)

Shelf Track – Relational Database & Inventory Management System

- Designed a relational database for inventory tracking with full CRUD and search, prioritizing data integrity and usability — foundations for reusable, trustworthy datasets for analytics and reporting.

Tech: Python, SQLite, OOP, Database Design, Data Visualization.

EDUCATION AND SELF STUDY

AI Engineering Bootcamp HyperionDev	02/2025 – 02/2026
Machine Learning Fundamentals Udemy	11/2025 – In Progress
Diploma in Pharmacy Technical Support (Pharmacy Technician) Nelson Mandela University	01/2017 – 11/2019

INTERESTS & TECHNICAL CURIOSITY

- Passionate about using data science and AI to improve assurance effectiveness and data-driven decision-making.
- Enjoys experimenting with analytics, machine learning, and NLP through hands-on projects.
- Actively explores emerging analytics tools and techniques to sharpen problem-solving and process efficiency.